

Hands-On: Conditional Rendering in React

In this lab, you will create a simple React component that conditionally renders content based on a boolean value in the component's state.

Step 1: Create a new React component

Create a new file called ``Greeting.js`` and add the following code:

```
import React, { Component } from 'react';

class Greeting extends Component {
  state = {
    isLoggedIn: false,
  };

  render() {
    const { isLoggedIn } = this.state;

    return (
      <div>
        {isLoggedIn ? (
          <h1>Welcome back!</h1>
        ) : (
          <h1>Please log in</h1>
        )}
        <button onClick={() => this.setState({ isLoggedIn: !isLoggedIn })}>
          {isLoggedIn ? 'Log out' : 'Log in'}
        </button>
      </div>
    );
  }
}

export default Greeting;
```

This component has an initial state with a boolean value to track whether the user is logged in or not. The ``render`` method conditionally renders a welcome message or a prompt to log in based on the ``isLoggedIn`` state. It also renders a button that updates the state when clicked to simulate logging in or out.

Step 2: Render the component

Create a new file called ``App.js`` and add the following code:

```
import React from 'react';
import Greeting from './Greeting';

function App() {
  return (
    <div className="App">
      <h1>Conditional Rendering in React</h1>
```

```
        <Greeting />
      </div>
    );
  }

  export default App;
```

This component simply renders the ``Greeting`` component.

Step 3: Run the app

Run the app using **npm start** and test the ``Greeting`` component by clicking the "Log in" button to see the welcome message and the "Log out" button to see the prompt to log in again.

Congratulations! You have successfully implemented conditional rendering in a React component.